TEAM MERAVIGLIA

ReLoop Inspector

Active visual quality assurance for the circular electronics economy

When vision is uncertain, ask for a better view.

An OpenCV 5 and AWS agent that guides evidence capture, proposes a cosmetic grade, and keeps the human responsible for the final decision.

FOCUS PATH	COUNTRY
Agentic Vision	Italy

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AWS Compute Grant Proposal · OpenCV AI Competition 2026

System architecture

OpenCV 5 perception · evidence orchestration · human approval · AWS audit trail

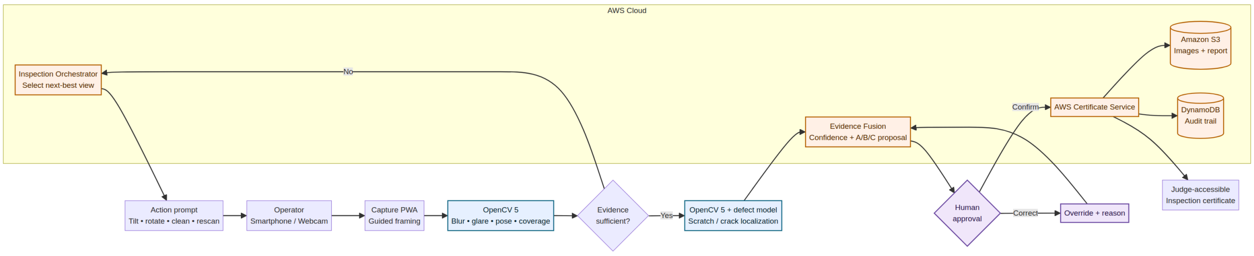

图 1 ReLoop Inspector turns visual uncertainty into a specific next action and an auditable decision.

Agentic Vision proof

OpenCV quality and coverage outputs determine whether the system can proceed. When evidence is insufficient, the orchestrator selects a new capture action; the next frame verifies whether the action improved the evidence.

1 Executive Summary

One-line proposition. ReLoop Inspector is an active visual agent that knows when the evidence is insufficient, asks the operator for the next best view, and produces a human-approved, auditable cosmetic grade for refurbished smartphones.

2 Problem and intended real-world impact

Cosmetic grading determines how a refurbished smartphone is reworked, priced and represented to the next buyer. Yet the decision is still heavily dependent on manual visual inspection. A 2025 open-access study in the *Journal of Remanufacturing* describes smartphone defect grading as labour-intensive and subjective, with disagreements and fatigue contributing to inconsistent decisions. The same study demonstrates that deep-learning computer vision can detect scratches and cracks and support faster, more consistent grading.[1]

The commercial problem is not merely detecting a visible mark. It is collecting sufficient, comparable evidence under variable lighting, viewpoints and device conditions. A single reflective image can turn a scratch into glare; a front-only scan can hide edge damage; two operators can apply the same grading policy differently. Industry practitioners also note that cosmetic grades vary across resale channels and that manual grading is inherently subjective.[2]

ReLoop Inspector addresses the part of the workflow that current classifiers usually ignore: **evidence acquisition**. The system inspects the quality and completeness of each view before attempting a final grade. If the visual evidence is ambiguous, it selects a specific next action — tilt, rotate, clean, move closer or capture an edge — and verifies the new evidence. Only then does it propose a transparent A/B/C grade for human approval.

The intended impact is to make the grading process more consistent, auditable and easier to learn, while preserving human authority. Better evidence can support fewer grading disputes, clearer buyer expectations, more targeted rework and longer device lifetimes. The competition prototype will test whether an agent-guided workflow improves evidence quality and human–system agreement compared with a single-shot baseline. It will not claim to certify hidden hardware faults, battery safety, water resistance or full device functionality.

3 Target users and beneficiaries

The primary users are technicians and quality-control operators in small and medium repair and refurbishment businesses. Operations managers benefit from a repeatable protocol and an audit trail. New operators benefit from in-context guidance rather than memorising a large inspection manual. Marketplace partners and consumers benefit indirectly from clearer and more traceable descriptions of device condition.

The project is deliberately designed as **decision support**, not worker replacement. The operator can confirm, reject or correct every proposed defect and grade. Overrides are recorded with reasons and become evidence for evaluation and future calibration.

4 The non-obvious insight

Most visual-quality systems optimise the prediction made from a fixed image. ReLoop optimises the **sequence of images needed to make a responsible decision**.

This reframes the problem from passive defect detection to active perception. When confidence is low, the correct output is not another confident label. It is a useful question: *What visual evidence is missing, and what action should the operator take next?* That behaviour creates a genuine perception–decision–action loop and aligns directly with the competition's Agentic Vision requirements.[3] [4]

5 Competition MVP and boundaries

The MVP will support one smartphone form factor and two exterior surfaces: the front screen and the rear cover. It will focus on two visible defect classes — scratches and cracks — and a transparent project-specific A/B/C cosmetic policy. This narrow scope is intentional: it enables rigorous testing, failure analysis and a working judge demonstration within the build window.

In scope	Explicitly out of scope
Image-quality checks: blur, glare, exposure and framing	Internal electronics and performance diagnostics
Device alignment and view-coverage tracking	Battery or electrical safety certification
Scratch and crack localisation	Universal grading across every marketplace
Agent-requested rescan and next-best-view guidance	Fully autonomous final decisions
Human-approved A/B/C proposal with evidence	Claims of legal or regulatory conformity
Cloud-backed report and audit trail	Production-scale throughput claims

6 Planned OpenCV 5 image and video analysis

OpenCV 5 will be a substantive runtime component throughout the workflow. The pipeline will use camera calibration, perspective correction, contour analysis, device registration, blur estimation, exposure and glare detection, frame selection, view-coverage tracking, defect-mask post-processing and visual overlays. A compact object-detection or segmentation model will localise scratches and cracks; OpenCV will convert model outputs into spatial evidence and interaction cues.

The system will maintain an **inspection evidence state** containing captured views, quality scores, surface coverage, defect candidates, uncertainty and human decisions. A deterministic policy will decide whether the evidence is sufficient. If it is not, the policy will request the next best view. The agent will then verify whether the operator completed the requested action before proceeding.

This design prevents an LLM from becoming the source of truth for grading. Language generation may explain an instruction or summarise evidence, but thresholds, state transitions and grade rules will remain explicit, testable and reproducible.

7 Planned AWS architecture and services

The capture interface will be a responsive web application usable from a phone or webcam. Lightweight quality checks may run locally for fast feedback. A meaningful cloud component will process evidence, maintain inspection state and generate the final certificate.

AWS component	Planned role
Amazon API Gateway	Secure entry point for inspection events and judge demo requests.
AWS Lambda or Amazon ECS	OpenCV inference orchestration, evidence-state transitions and report generation.
Amazon S3	Storage of consented inspection images, masks, evaluation artefacts and certificates.
Amazon DynamoDB	Audit trail for actions, confidence values, overrides and session state.
Amazon CloudWatch	Latency, failures, rescan events and system observability.
Amazon Bedrock, optional	Constrained explanation of structured evidence and operator guidance; not final grading authority.

The deployment will use pinned dependencies, environment-based configuration, least-privilege access and a reproducible judge setup. Personally identifying content, serial numbers and unrelated screen content will be removed or blurred before persistence.

8 Agentic Vision loop

Stage	System behaviour	Competition evidence
Perception	OpenCV evaluates quality, pose, coverage and candidate defects.	Frame overlays, masks and numeric quality trace.
Decision	The orchestrator determines whether the evidence meets an explicit sufficiency policy.	State transition log and confidence rationale.
Action	The system selects and displays a specific next capture instruction.	Recorded prompt: tilt, rotate, clean, zoom or edge view.
Verification	OpenCV verifies that the requested view or quality improvement occurred.	Before/after quality and coverage comparison.

Stage	System behaviour	Competition evidence
Human control	The operator confirms or corrects the proposed defect and grade.	Override reason and final approval event.
Outcome	AWS generates an evidence-backed inspection certificate.	Judge-accessible report and audit trail.

9 Data strategy

The research literature demonstrates technical feasibility but also highlights the scarcity and specificity of industrial defect data.1 The paper's public page does not provide a reusable dataset. We will therefore build a small, controlled competition dataset rather than imply access to proprietary industry data.

The dataset plan contains three layers. First, the team will collect owned images of used smartphones under controlled variations in lighting, angle, distance, cleanliness and surface condition. Second, the team will use licence-compatible public crack and surface-anomaly data only for pre-training or method exploration, documenting provenance and licence terms. Third, synthetic augmentation will reproduce glare, blur, perspective shifts and contrast variation, while the final evaluation will remain on real, held-out images.

Two human graders will independently label defect location and severity according to the project policy. Disagreements will be adjudicated and retained as a measure of ambiguity rather than removed from the record.

10 Evaluation method

The central experiment compares a **single-shot baseline** with the **agent-guided acquisition workflow**. Both conditions use the same defect model and grading policy. The difference is whether the system may reject insufficient evidence and request a targeted rescan.

Metric	Definition	Decision value
Defect precision and recall	Per-class detection performance for scratches and cracks.	Measures core technical execution.
Human–agent agreement	Agreement with the adjudicated two-grader result.	Tests grading consistency.
Evidence recovery rate	Share of low-confidence cases made usable after an agent-requested rescan.	Measures Agentic Vision value.
Inspection completion rate	Sessions producing a reviewed certificate within two rescans.	Measures task effectiveness.
Median time to certificate	Time from first capture to final human approval.	Measures operational usability.

Metric	Definition	Decision value
Override rate and reason	Frequency and cause of human correction.	Measures failure handling and human control.
End-to-end latency	Time for each cloud decision and report step.	Measures AWS delivery quality.

No target accuracy will be claimed before data collection. The project will report confidence intervals where sample size allows, disclose class imbalance and show representative failure cases. The published research baseline — including its reported 70.4% precision and 8 ms YOLOv8x inference result — will be treated as context, not as a performance claim for ReLoop.[1]

11 Judge demonstration

The five-minute demonstration will open with two operators assigning different grades to the same device. ReLoop will then receive a reflective first scan and refuse to produce a grade. The interface will request a precise tilt and edge view; OpenCV will verify the new evidence and localise the defect. The system will propose Grade B with a visible confidence rationale. The operator will correct one ambiguous label, and the override will be stored. AWS will then generate an inspection certificate and audit trail.

The final screen will compare the single-shot and agent-guided conditions, showing whether the rescan improved evidence quality, reduced uncertainty and changed the outcome. A failure case will be included deliberately to demonstrate responsible operation.

12 Innovation and differentiation

ReLoop is not another object-detection demo. Its distinctive contribution is the combination of active perception, explicit evidence sufficiency, targeted human action, verification and accountable approval. The agent is useful precisely because it can decline to decide.

The concept also connects technical depth to a credible adoption path. It starts with a bounded task that can be tested on ordinary hardware, creates auditable value for operators and can later expand to additional surfaces, device families and rework recommendations. The MVP remains valuable even without full automation because it standardises evidence capture and documents decisions.

13 Responsible operation and limitations

The team will use only owned or appropriately licensed images, models and media. The application will remove personal screen content and serial identifiers before cloud storage. Raw images will use retention limits, and the demo dataset will contain no personal data.

The system will display uncertainty and will require human approval for every final grade. It will not imply that a cosmetic result certifies device safety or function. Known risks include reflections mistaken for scratches, fine defects below camera resolution, protective films, out-of-distribution device shapes, inconsistent human labels and operator distrust. Each risk will have a test, a visible limitation statement and a mitigation path.

14 Team strength and roles

Team MERAVIGLIA brings together more than thirty-five combined years of strategic design, business design, marketing, facilitation and organisational transformation. Both members are co-founders of Marketing Toys, an Italian benefit corporation that works with companies, startups, industrial organisations, cultural institutions, territories and public bodies.5 6 7

Filippo Giustini — Design Strategy Lead. Filippo is a Design Strategist with more than fifteen years of experience helping organisations and territories design sustainable transformations and value systems. His work combines business design, co-design, positioning, facilitation and storytelling. He will lead system framing, product strategy, operator experience, evaluation logic, coordination and the final narrative.5 6

Gaia Provvedi — Business Design & Validation Lead. Gaia is a Business Designer and Marketing Advisor with more than twenty years of experience managing teams and strategic communication and marketing projects. Her work focuses on sustainable growth, value propositions, circular business models, benefit corporations and Design Thinking. She will lead user discovery, assumptions testing, pilot design, adoption strategy, business modelling and ethical impact.5 7

The team does not present unsupported prior computer-vision or AWS credentials. Its technical credibility will be demonstrated through a narrow MVP, an explicit architecture, reproducible tests, documented use of OpenCV 5 and AWS, and working evidence delivered during the competition. AI-assisted coding is permitted, but the team will review, test and explain all submitted code.3 4

15 Eight-week delivery plan

Week	Deliverable	Acceptance gate
1	Five practitioner interviews, grading policy, dataset consent and capture protocol.	Validated pain, clear scope, two test devices.
2	Controlled dataset v1 and OpenCV quality-check baseline.	Blur, glare, pose and framing traces run locally.
3	Device registration, view coverage and defect baseline.	Reproducible front/rear capture and masks.
4	Evidence state, next-best-view policy and AWS event pipeline.	Visual output triggers a new capture action.
5	Responsive PWA, human review and override workflow.	End-to-end certificate from a real device.
6	Baseline comparison, error analysis and UX study.	Metrics table and at least five documented failures.
7	Reliability, observability, security and judge rehearsal.	Stable demo, pinned build and recovery procedure.
8	Technical report, repository, video and Devpost submission.	Every official requirement checked twice.

16 Use of the AWS compute grant

The requested grant will be used for containerised inference experiments, storage of the consented evaluation set, test deployments, observability and repeated judge-ready builds. It will not be used for unrelated workloads. The team will attend the required midpoint check-in and provide a concise progress update with working traces and metrics.

17 Why this team, why now

The transition to a circular electronics economy will not be achieved by models alone. It requires trustworthy workflows that fit real decisions, make uncertainty visible and help people act consistently. Team MERAVIGLIA's strength is designing that connection between technology, behaviour and sustainable value.

ReLoop Inspector begins with a small but consequential question: *When the camera has not seen enough, can the system ask for the right evidence instead of pretending to know?* The project will answer that question with a working OpenCV 5 and AWS prototype, measurable results and a responsible human-in-the-loop design.

18 References